

Intermediate Microeconomics — Week 2, Day 2

In-class Exercises: Welfare, Price Controls, and Tax Incidence

Perloff, Chs. 8–9 • work in pairs • solutions at the end

Exercise 1 — Surplus and a binding price ceiling [competency: measure welfare, evaluate a control]

A city rental-services market is described by

$$Q_d = 100 - 2P, \quad Q_s = -20 + 2P,$$

with P the monthly price (\$) and Q in thousands of contracts.

- (a) Find the equilibrium price and quantity.
 - (b) Compute consumer surplus, producer surplus, and total surplus. (Hint: write the inverse curves first.)
 - (c) The council imposes a price ceiling of $\bar{P} = 20$. Find the quantity actually traded, the size of the shortage, and the deadweight loss.
 - (d) Explain, in surplus terms, who is helped and who is harmed. Under *efficient* rationing, does aggregate consumer surplus rise or fall relative to (b)?
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Exercise 2 — Tax incidence [competency: incidence & elasticities, not surplus geometry]

A market for a consumer good is

$$Q_d = 60 - P, \quad Q_s = -30 + 2P.$$

The government introduces a specific tax of $t = 6$ per unit.

- (a) Find the pre-tax equilibrium, and the price elasticities of demand and supply *at that equilibrium*.
 - (b) Suppose the tax is legally collected from **buyers**. Find the price buyers pay P_d , the price sellers receive P_s , the quantity traded, and the per-unit burden borne by each side.
 - (c) Show that the consumer share of the burden equals $\frac{\eta_s}{\eta_s + |\varepsilon_d|}$. Which side bears more, and why?
 - (d) Compute tax revenue and the deadweight loss.
 - (e) If instead the tax were legally collected from **sellers**, would the burden split change? Demonstrate.
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Reconvene: for Exercise 1 be ready to state the DWL and *why* it arises; for Exercise 2, state the incidence split and the one number (an elasticity ratio) that drives it.